

ArcGIS Maps SDK for JavaScript: App Development with Components, Part 4: Extending and Styling



Kitty Hurley



Rene Rubalcava

```
Map map = new Map(BasemapStyle.ArcGISNavigation);  
  
map.InitialViewpoint = new Viewpoint(40.72,  
-73.995,  
72000);  
  
mapView.Map = map;
```

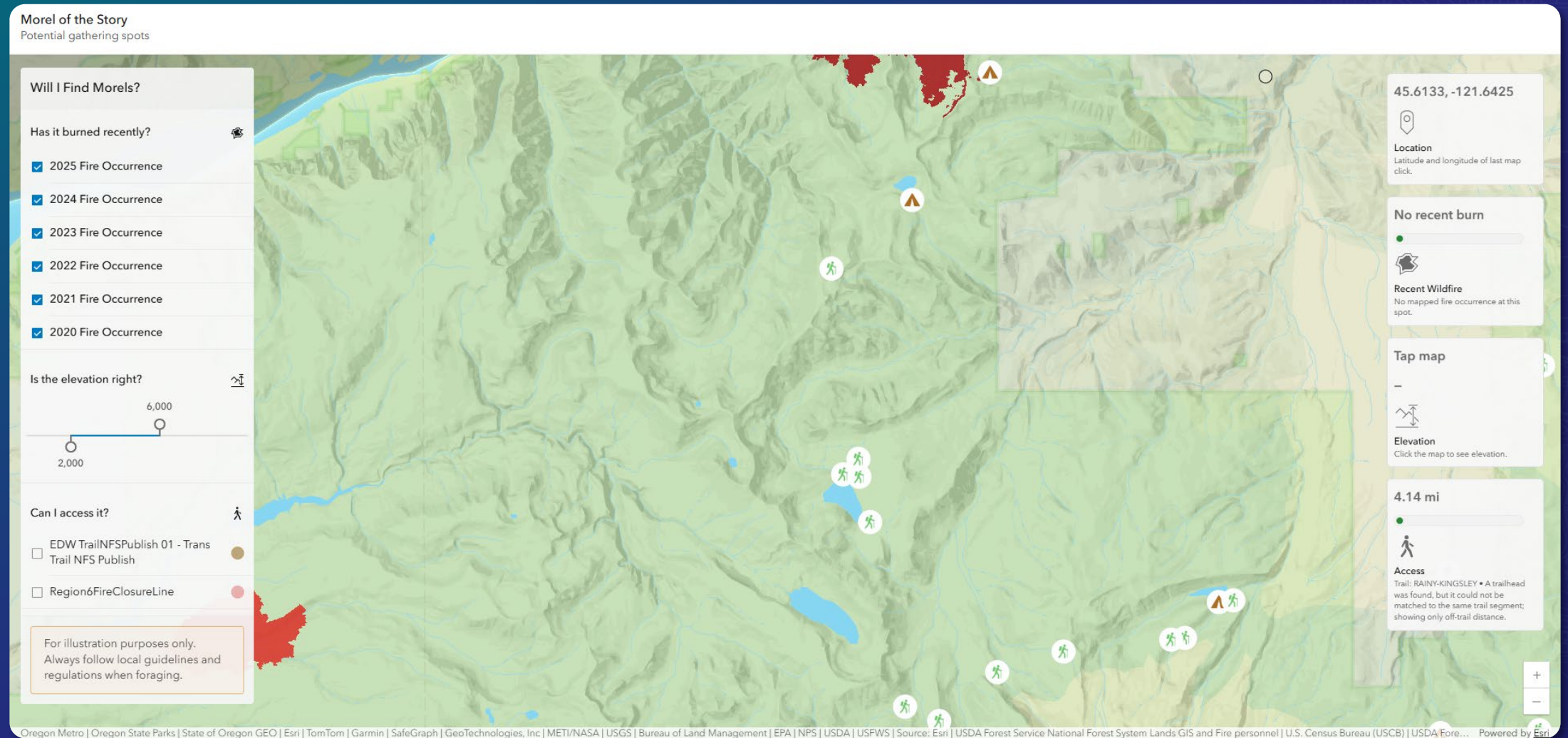
```
class MainActivity : AppCompatActivity() {  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        setContentView(  
            SampleAppTheme {  
  
                val map = ArcGISNavigationMap()  
  
                map.InitialViewpoint = Viewpoint(  
                    40.72,  
                    -73.995,  
                    72000.0  
                )  
            }  
        )  
    }  
}
```

**DEV
&
TECH**

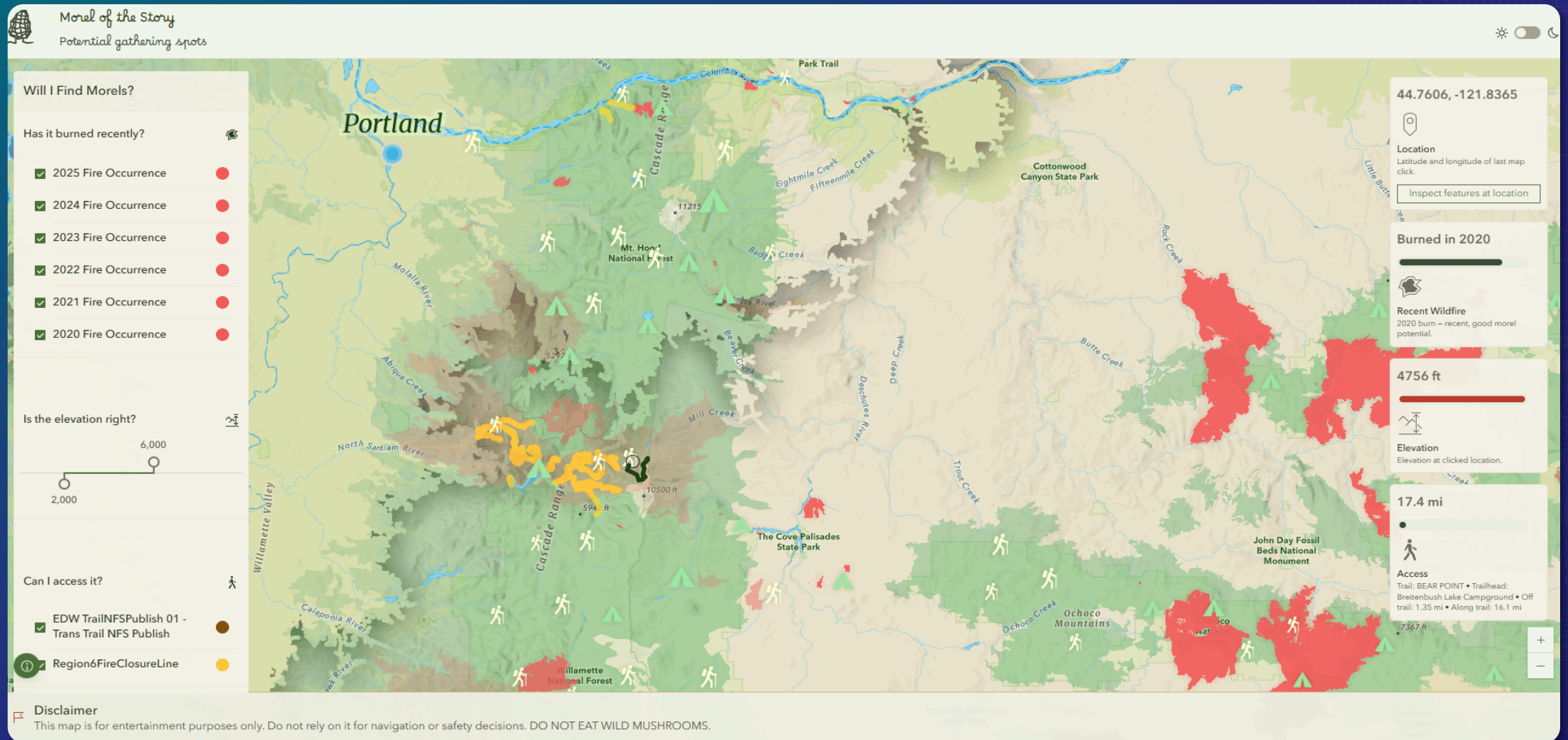
Agenda

- Introduction
- Modes
- Design tokens
 - Semantic tokens
 - Component tokens
- Application customization
- Discussion and resources

Our application



Theming an application



Modes

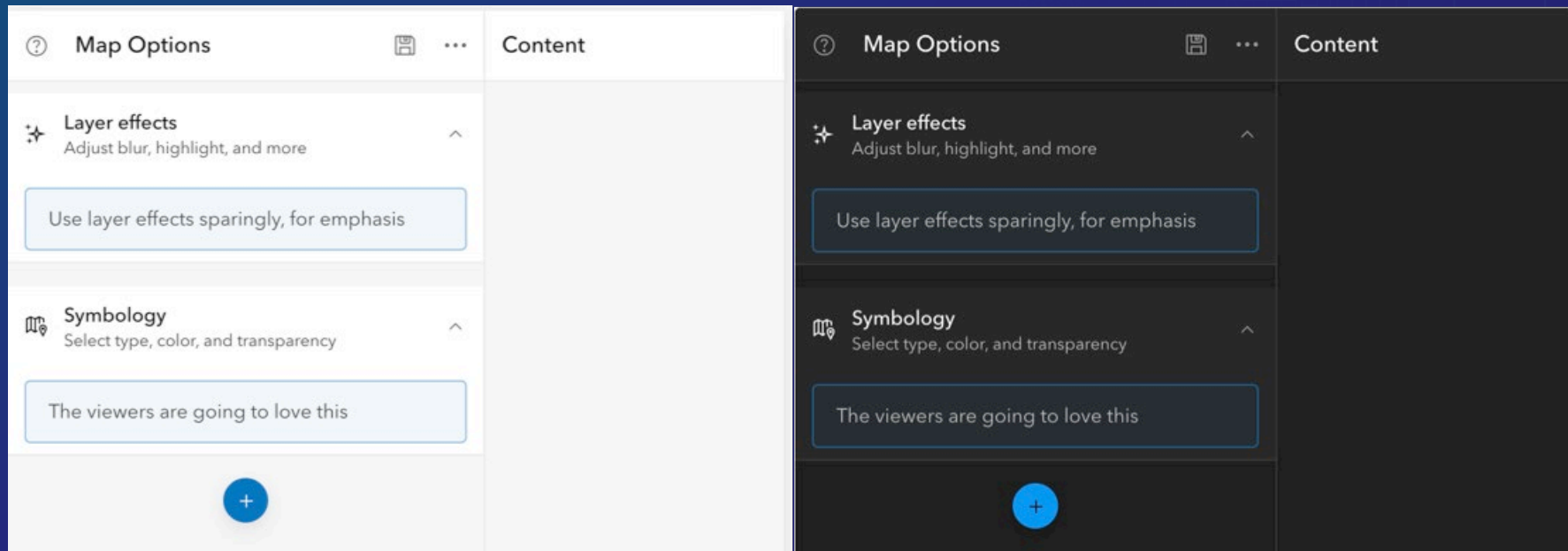
Designing for light and dark modes



Calcite modes

<https://developers.arcgis.com/calcite-design-system/core-concepts/#modes>

- **Auto** `--calcite-mode-auto`
- **Light** `--calcite-mode-light`
- **Dark** `--calcite-mode-dark`



How to set modes

- Light and Dark mode are flexible

```
<html class="calcite-mode-light"></html>
<body class="calcite-mode-light"></body>
<calcite-shell content-behind class="calcite-mode-light"></calcite-shell>
<calcite-shell content-behind class="calcite-mode-dark"></calcite-shell>

<!-- You could even change specific components -->
<arcgis-legend class="calcite-mode-dark"></arcgis-legend>
```


How to set modes

<https://developers.arcgis.com/calcite-design-system/tutorials/build-a-dark-mode-switch>

```
<calcite-shell content-behind className={`calcite-mode-${mode}`}>
  {content}
</calcite-shell>
```

```
let darkModeDisabled = true;
const updateDarkMde = ( ) => {
  darkModeDisabled = !darkModeDisabled;
  document.getElementById( "app-loader" ).hidden = false;
  // Calcite mode
  document.body.classList.toggle( "calcite-mode-dark" );
};
```


How to set modes

- Be mindful of `prefers-color-scheme`

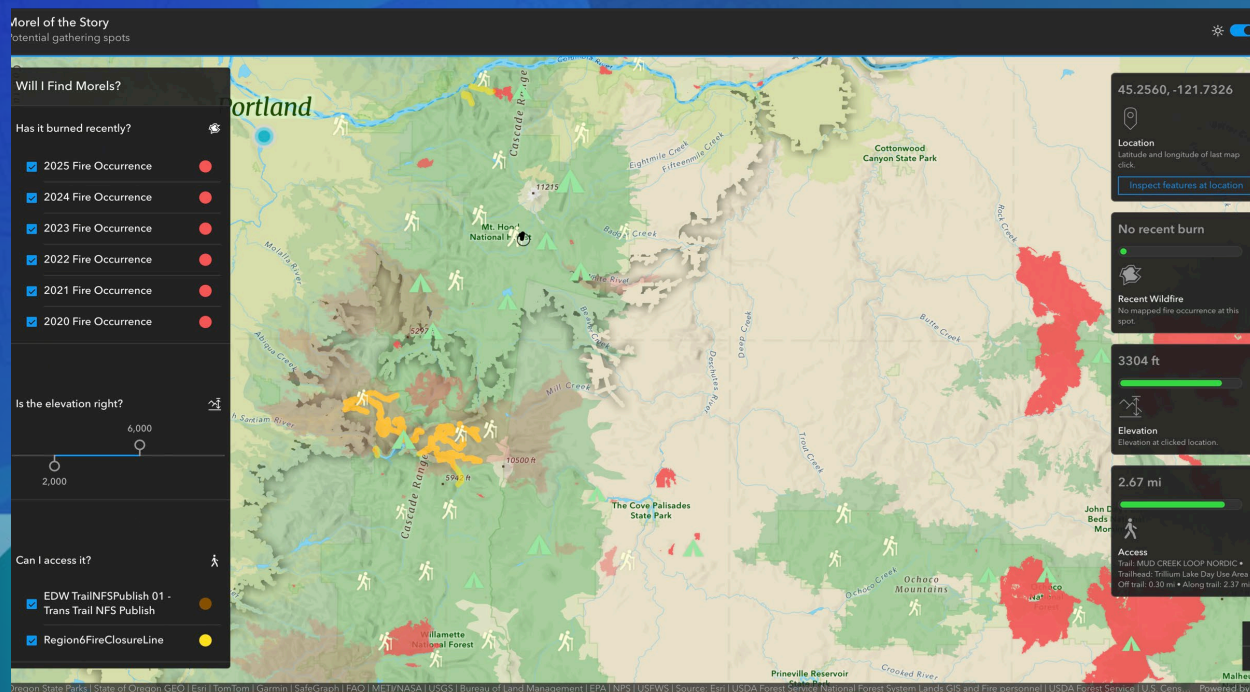
```
function getInitialThemeMode(): ThemeMode {  
    if (typeof window === "undefined") {  
        return "light";  
    }  
  
    return window.matchMedia("(prefers-color-scheme:  
dark)").matches  
        ? "dark"  
        : "light";  
}
```

Attribution mode

<https://developers.arcgis.com/javascript/latest/references/map-components/components/arcgis-map/#properties>

- Attribution is not a web component
 - It's part of the underlying view
 - Doesn't see calcite modes

Property	Attribute	Type
<code>allLayerViews</code> readonly		<code>ReadonlyCollection<LayerView></code>
<code>analyses</code>		<code>Collection<Analysis></code>
<code>animationsDisabled</code>	<code>animations-disabled</code>	<code>boolean</code>
<code>aria</code>		<code>MapView["aria"]</code>
<code>attributionItems</code> readonly		<code>MapView["attributionItems"]</code>
<code>attributionMode</code>	<code>attribution-mode</code>	<code>"dark" "light" null undefined</code>
<code>autoDestroyDisabled</code>	<code>auto-destroy-disabled</code>	<code>boolean</code>
<code>background</code>		<code>ColorBackground null undefined</code>
<code>basemap</code>	<code>basemap</code>	<code>Basemap string undefined</code>
<code>basemapView</code>		<code>BasemapView<LayerView></code>
<code>canZoomIn</code> readonly		<code>boolean</code>
<code>canZoomOut</code> readonly		<code>boolean</code>
<code>center</code>	<code>center</code>	<code>MapView["center"]</code>



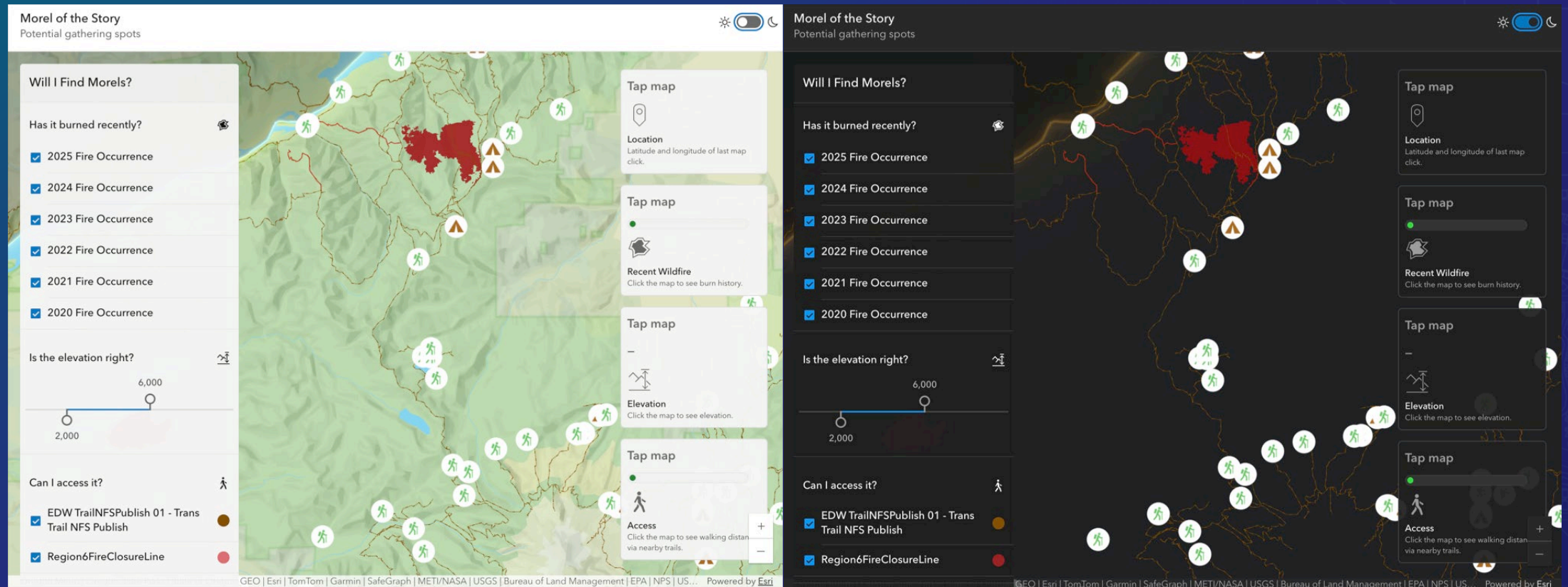
Demo:

Switching Modes

René Rubalcava

Compliment modes with Basemap

- Switch the basemap when switching themes



Compliment modes with Basemap

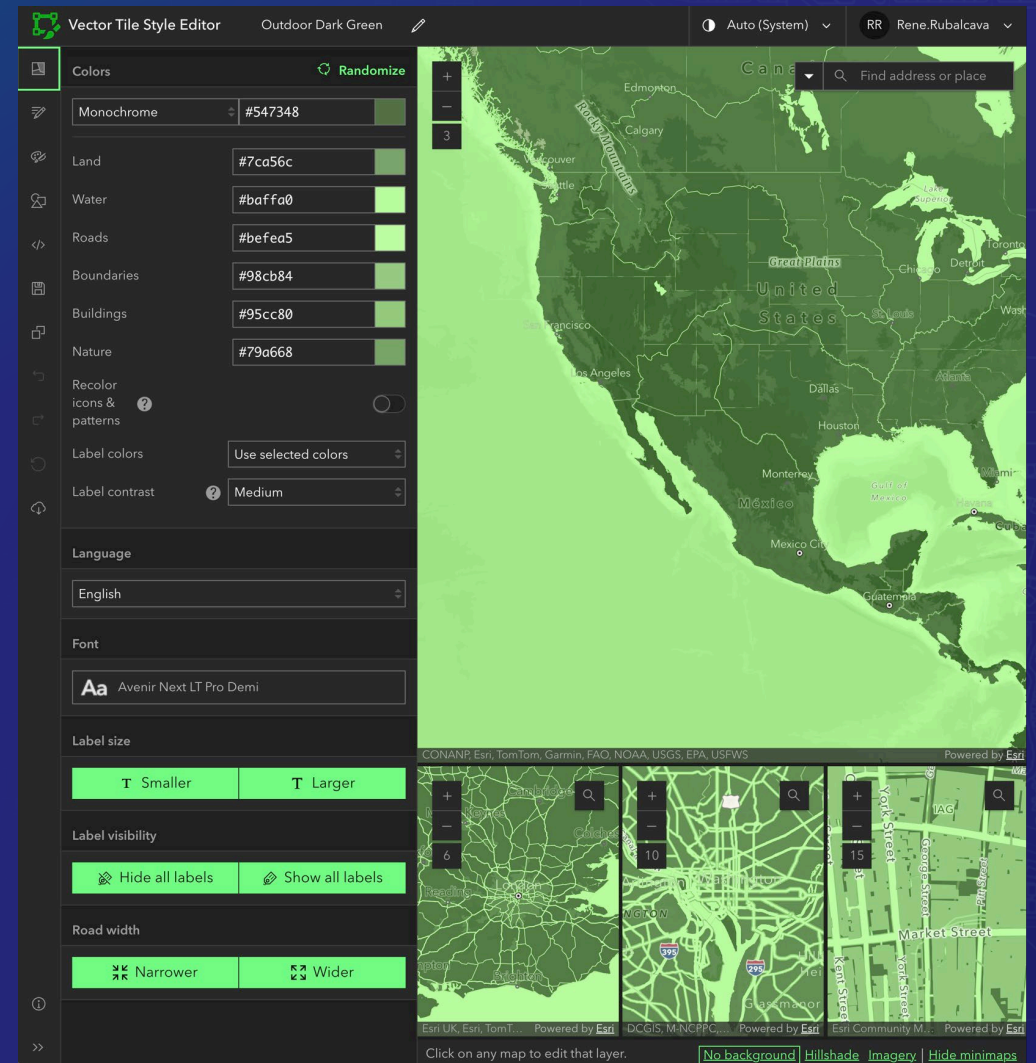
- Switch the basemap when switching themes

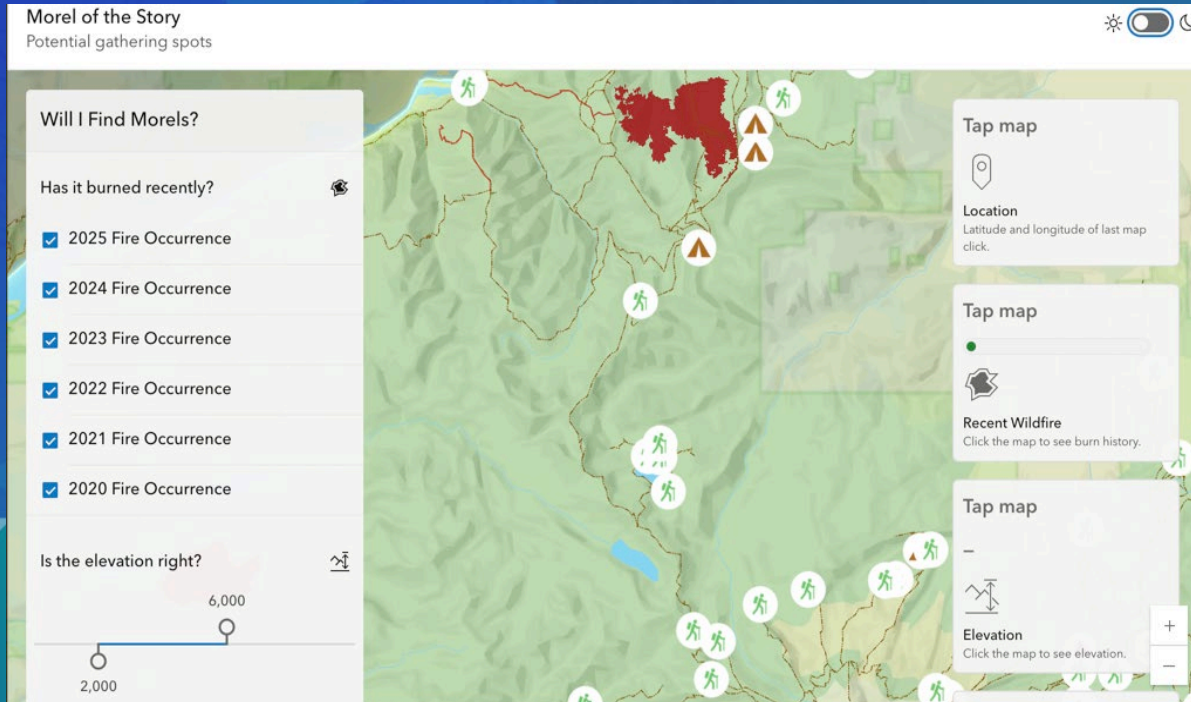
```
const darkBackgroundLayer = new VectorTileLayer({
  url: darkStyleUrl,
  title: "Dark basemap",
  visible: true,
});

const groupLayer = map.layers.find(
  (layer) => layer.title === "BG",
) as GroupLayer;
const index = groupLayer.layers.findIndex(
  (layer) => layer.title === backgroundLayer?.title,
);
groupLayer.layers.splice(
  index,
  1,
  isDarkMode ? darkBackgroundLayer : backgroundLayer,
);
```

Compliment modes with Basemap

- Use a custom basemap to match your theme
- Allows extra branding
- Be mindful of choices





Demo: Basemap modes

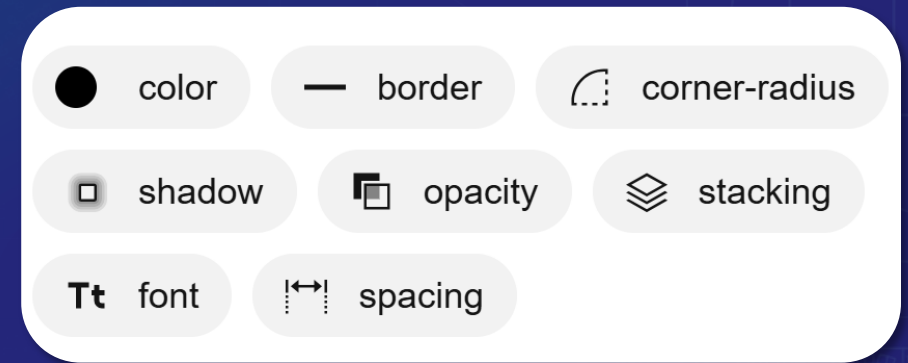
Design tokens

Building blocks of design



What are design tokens?

- Represent decisions (e.g., color and spacing)
- Encourage consistency
- Reinforce accessibility
- Support rich theming and customization



Design token tiers

- **Core tokens**

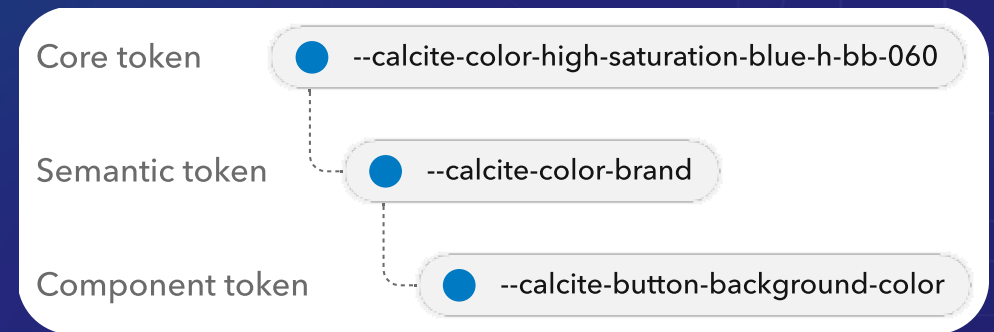
- Variables used as values for semantic tokens
- Predetermined raw values are not intended for apps

- **Semantic tokens**

- Use core tokens with purpose-based naming
- Reference core tokens instead of raw values

- **Component tokens**

- Component-level overrides with customization
- Scoped to respective components
- Reference semantic tokens



Customization with semantic tokens



Semantic tokens

```
:root {  
  --calcite-color-text-1: #3c83b7;  
  --calcite-color-text-2: #3c83b7;  
  --calcite-color-text-3: #3c83b7;  
}
```




Demo: Global styling

Using semantic tokens

<https://developers.arcgis.com/calcite-design-system/foundations/tokens/reference/#semantic-tokens>

esri Developer

DocumentationFeatures ▾SDKs and APIs ▾Products ▾Support ▾

Calcite Design System / Foundations

HomeFoundationsComponentsResourcesIcons

Overview

Accessibility

Design tokens

- Usage
- Reference

Colors and theming

Typography

Iconography

Layout patterns

Design token reference

Current version: [@esri/calcite-design-tokens: 4.0](#) (February 2026)

Semantic tokens

```
@import "@esri/calcite-design-tokens/dist/css/index.css";
```

Colors

Name	Light value	Dark value	Platform tokens
Surface			
1	 #f7f7f7	 #212121	<code>var(--calcite-color-surface-1)</code>
2	 #ffffff	 #2b2b2b	<code>var(--calcite-color-surface-2)</code>
3	 #f2f2f2	 #363636	<code>var(--calcite-color-surface-3)</code>
4	 #ebebeb	 #404040	<code>var(--calcite-color-surface-4)</code>
highlight	 #d5e9ff	 #2b465f	<code>var(--calcite-color-surface-highlight)</code>

On this page

Semantic tokens

- Colors
- Border
- Corner radius
- Shadow
- Opacity
- Stacking
- Font
- Spacing

Core tokens

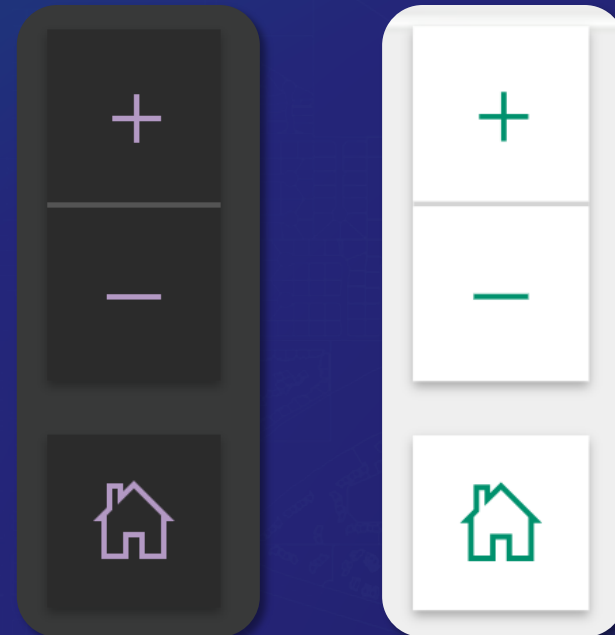
- Colors
- Shadow
- Opacity
- Stacking

Customization with component tokens



Component tokens

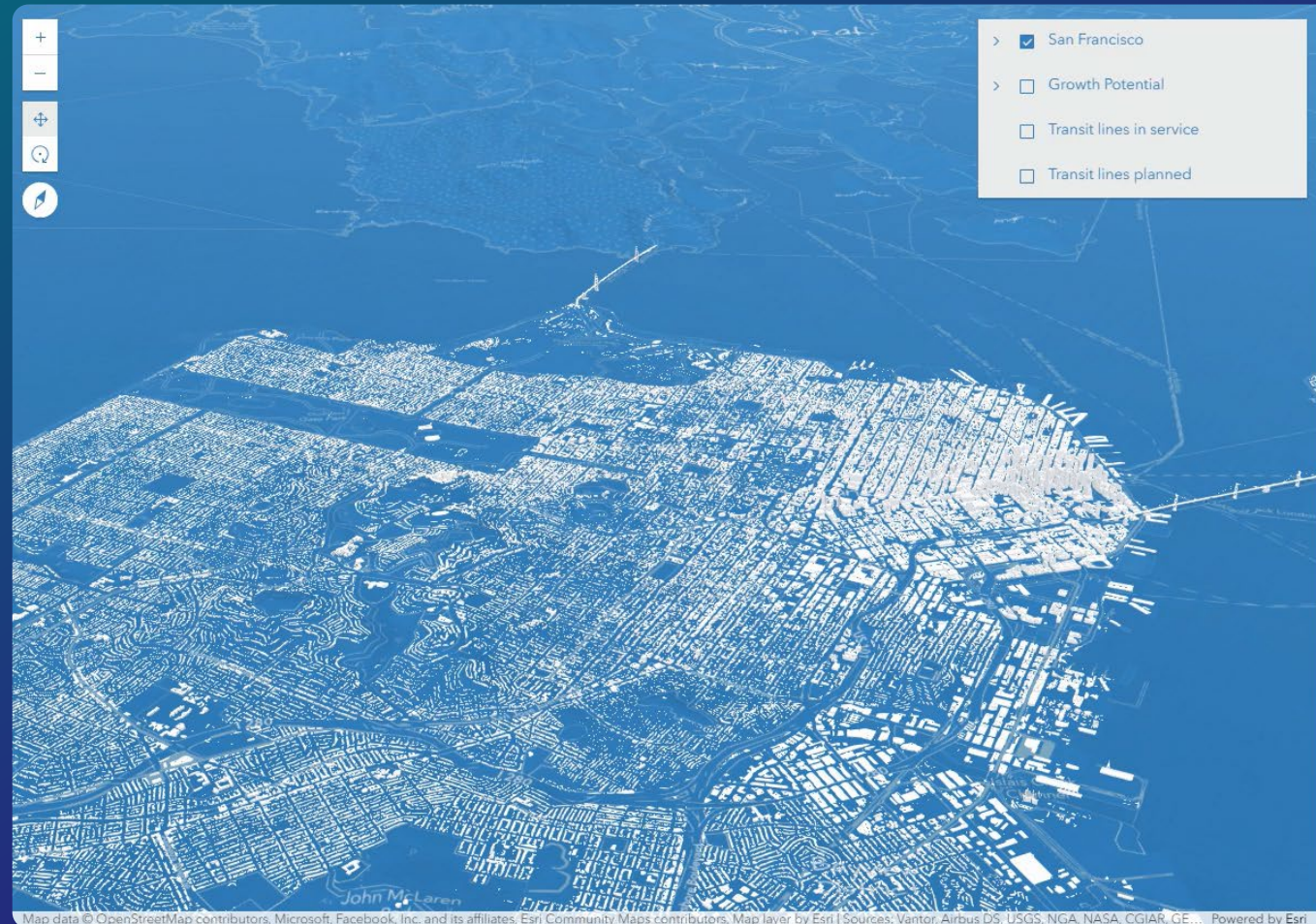
- Component-level overrides with customization
- Scoped to respective components
- Reference semantic tokens
- ArcGIS Maps SDK for JavaScript Map components
 - Use Calcite as their foundational building blocks
 - Can leverage Calcite component tokens for styling



Inspect components further

<https://developers.arcgis.com/calcite-design-system/components/list-item/#styles>

<https://codepen.io/geospatalem/pen/OPRNLbx>



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```
<script type="module" src="https://js.arcgis.com/5.0/"></script>
<arcgis-scene item-id="adfad6ee6c6043238ea64e121bb6429a" hydrated>
  #shadow-root (open)
  <arcgis-zoom slot="top-left" layout="vertical" hydrated>
    #shadow-root (open)
  <arcgis-navigation-toggle slot="top-left" layout="vertical" navigation-mode="pan" state="disabled" hydrated>
    #shadow-root (open)
  <arcgis-compass slot="top-left" visual-scale="m" icon="compass-needle" hydrated>
    #shadow-root (open)
  <arcgis-layer-list slot="top-right" visibility-appearance="checkbox" hydrated>
    #shadow-root (open)
    #adopted-style-sheet
    #adopted-style-sheet
    <div arcgis-widget-wrapped class="esri-layer-list esri-widget esri-widget-panel">
      #shadow-root (open)
      #calcite-flow calcite-hydrated
        #shadow-root (open)
        #calcite-flow-item overlay-positioning="absolute" scale="m" selected heading-level="2" calcite-hydrated
          #shadow-root (open)
          #calcite-list data-layer-type="operational-19cc682aabe-object-0" filter-text interaction-mode="interactive" display-mode="nested" scale="m" selection-appearance="border" selection-mode="none" filter-placeholder group="operational-19cc682aabe-object-0" calcite-hydrated
```

logged-in div#result-iframe-wrap iframe#result.result-iframe html body arcgis-scene arcgis-layer-list

Styles Computed Layout Event Listeners DOM Breakpoints Properties Accessibility State

Filter

Console AI assistance What's new Issues Search Animations

OPRNLbx

Default levels 4 Issues

ctrl i to turn on code suggestions. Don't show again

Component tokens in use

<https://developers.arcgis.com/calcite-design-system/components/action/#styles>

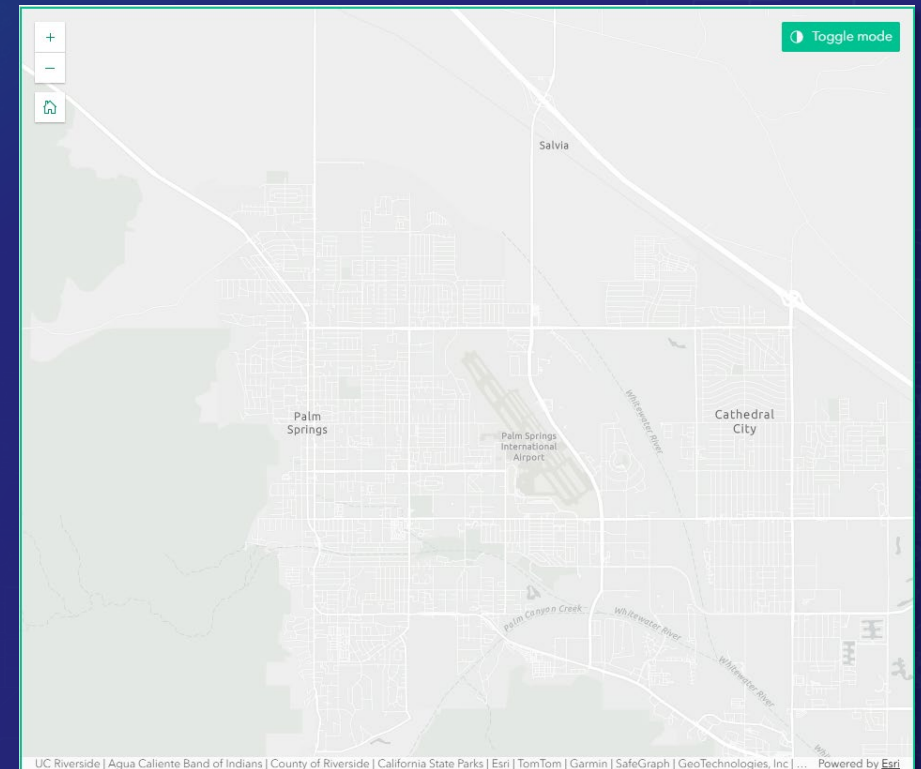
<https://codepen.io/geospatialempen/gbMVVMB>

```
body {
  --calcite-color-brand: #00c291;
  --calcite-color-brand-hover: #00ad82;
  --calcite-color-brand-press: #00916d;
  --calcite-color-focus: var(--calcite-color-brand);
}

body.calcite-mode-dark {
  --calcite-color-brand: #d6b9eb;
  --calcite-color-brand-hover: #c59cd6;
  --calcite-color-brand-press: #b399c4;
  --calcite-color-focus: var(--calcite-color-brand);
}

#toggle-mode {
  --calcite-action-text-color: var(--calcite-color-surface-1);
  --calcite-action-text-color-press: var(--calcite-color-surface-4);
  --calcite-action-background-color: var(--calcite-color-brand);
  --calcite-action-background-color-press: var(--calcite-color-brand-press);
  --calcite-action-background-color-hover: var(--calcite-color-brand-hover);
}

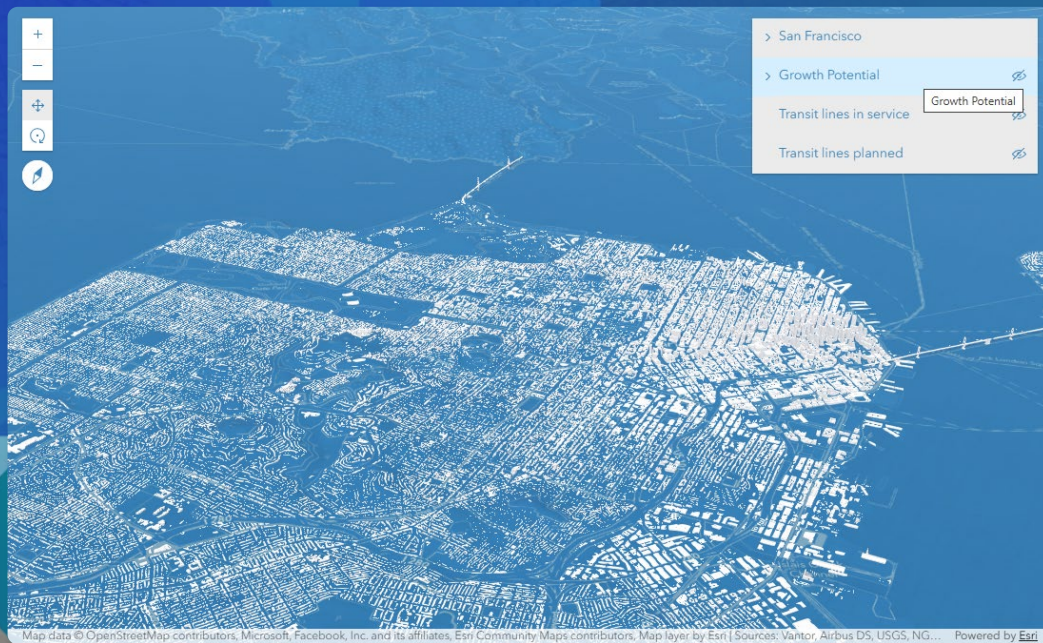
arcgis-zoom,
arcgis-home {
  --calcite-button-icon-color: var(--calcite-color-brand-press);
}
```



Semantic and component tokens

<https://codepen.io/geospatialempen/pen/OPRNLbx>

```
:root {  
  --calcite-color-text-1: #3c83b7;  
  --calcite-color-text-2: #3c83b7;  
  --calcite-color-text-3: #3c83b7;  
}  
  
arcgis-layer-list {  
  --calcite-list-background-color: var(--calcite-color-surface-4);  
  --calcite-list-background-color-hover: var(--calcite-color-surface-highlight);  
}
```



Demo:

Component tokens

Kitty Hurley

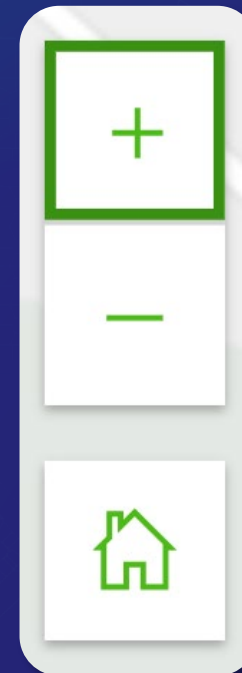
Create your own variables

<https://codepen.io/geospatialempen/jEMqPvd>

```
:root {
  --kellyGreen: #4CBB17;
  --lighterGreen: #5BE31C;
  --darkerGreen: #3A9112;
}

body {
  --calcite-color-brand: var(--kellyGreen);
  --calcite-color-brand-hover: var(--lighterGreen);
  --calcite-color-brand-press: var(--kellyGreen);
  --calcite-color-focus: var(--calcite-color-brand);
}

arcgis-zoom,
arcgis-home {
  --calcite-button-icon-color: var(--calcite-color-brand);
  --calcite-color-focus: var(--darkerGreen);
  --calcite-offset-invert-focus: 1;
}
```



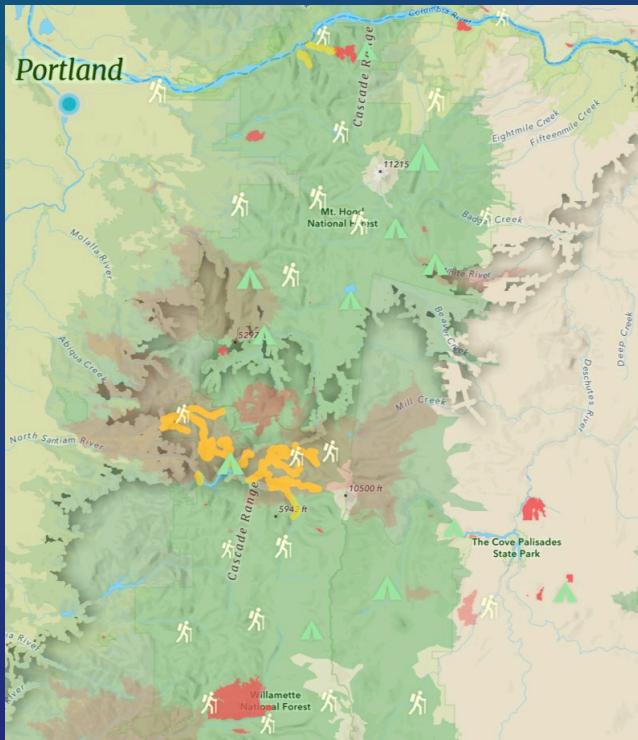
Application customization

Adding flair



Brand tokens to map

- Light and dark modes
- AI generated (ssshhhh)



- "Here are some Calcite tokens for light and dark theme"
- "Here is an image of my map"
- "Create alternate tokens that compliment this map"
- Tweak the results

Choosing a font

- Align with your brand
- Use `font-family` inheritance
- Ensure fallback fonts are available
- Prioritize readability

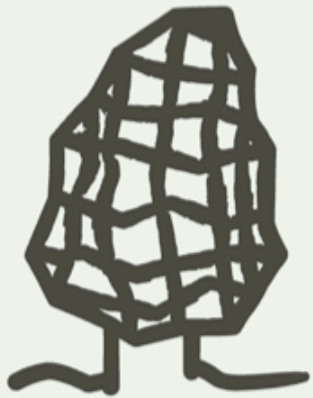


Morel of the Story

```
:root {  
  font-family: "Borel", Arial, sans-serif;  
}
```


Logo

- Woo!
- Animations
- SVG Logos work great!
- Favicon



Morel of the Story
Potential gathering spots

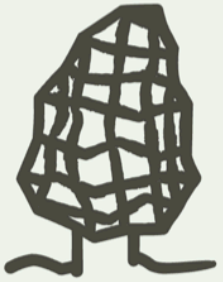
Logo

- SVG path colors using `fill="someColor"`
- Don't need to target path with css

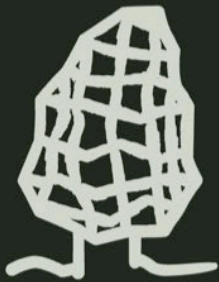
```
<svg className="logo" width="48" height="48" viewBox="0 0 48 48"
fill="none" xmlns="http://www.w3.org/2000/svg">
  <path className="top" d="..." fill="currentColor" />
  <path className="bottom" fill-rule="evenodd" clip-rule="evenodd"
d="..." fill="currentColor" />
</svg>
```

Logo

- Inherits color from parent



Morel of the Story
Potential gathering spots



Morel of the Story
Potential gathering spots

Logo

- Make your logos dance!

```

● ● ●

.logo:hover {
  cursor: pointer;
}

.logo:hover svg {
  animation: wiggle-rotate 0.5s ease-in-out infinite
  alternate; transform-origin: center;
}

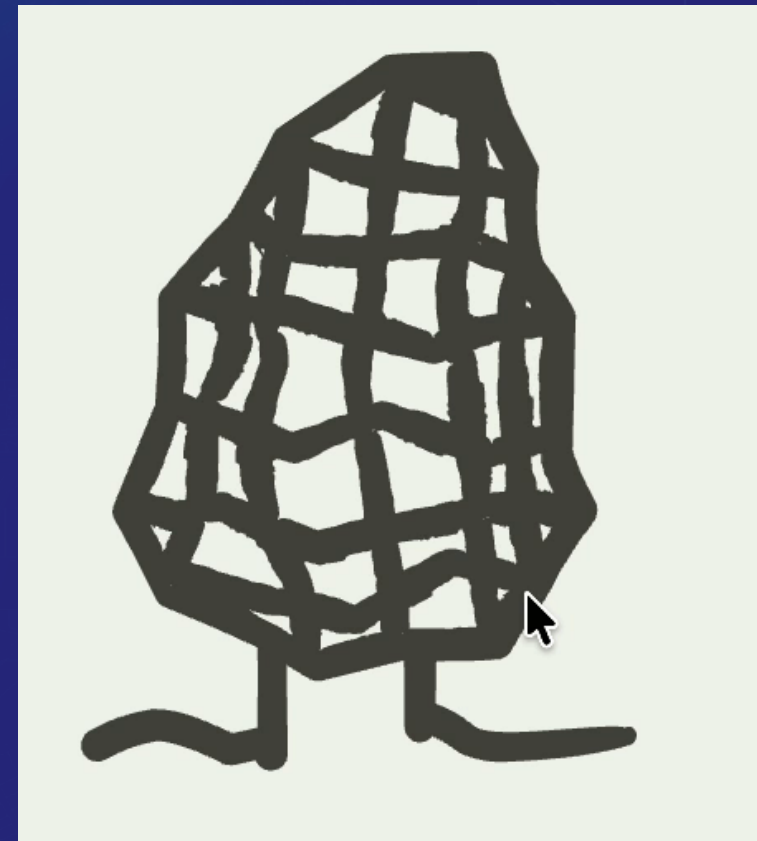
.logo:hover .top,
.logo:hover .bottom {
  transform-box: fill-box;
  transform-origin: center;
  animation: wiggle-skew 0.6s ease-in-out infinite alternate;
}

/* Reverse just the bottom */
.logo:hover .bottom {
  animation-direction: alternate-reverse;
}

/* Shared rotation */
@keyframes wiggle-rotate {
  from {
    transform: rotate(-4deg);
  }
  to {
    transform: rotate(4deg);
  }
}

/* Shared skew */
@keyframes wiggle-skew {
  from {
    transform: skewX(-6deg);
  }
  to {
    transform: skewX(6deg);
  }
}

```





Morel of the Story

Potential gathering spots

Will I Find Morels?

Has it burned recently?



☒ 2025 Fire Occurrence



☒ 2024 Fire Occurrence



☒ 2023 Fire Occurrence



Demo: Fonts and Logo

René Rubalcava

Attribution customization

- Customize attribution messaging/icons
- Padding customization
- Sizing customization
- Alter based on the UI changes (expansion/collapse)

Attribution customization

<https://developers.arcgis.com/javascript/latest/references/map-components/components/arcgis-map/#styles>

- Available tokens

Styles

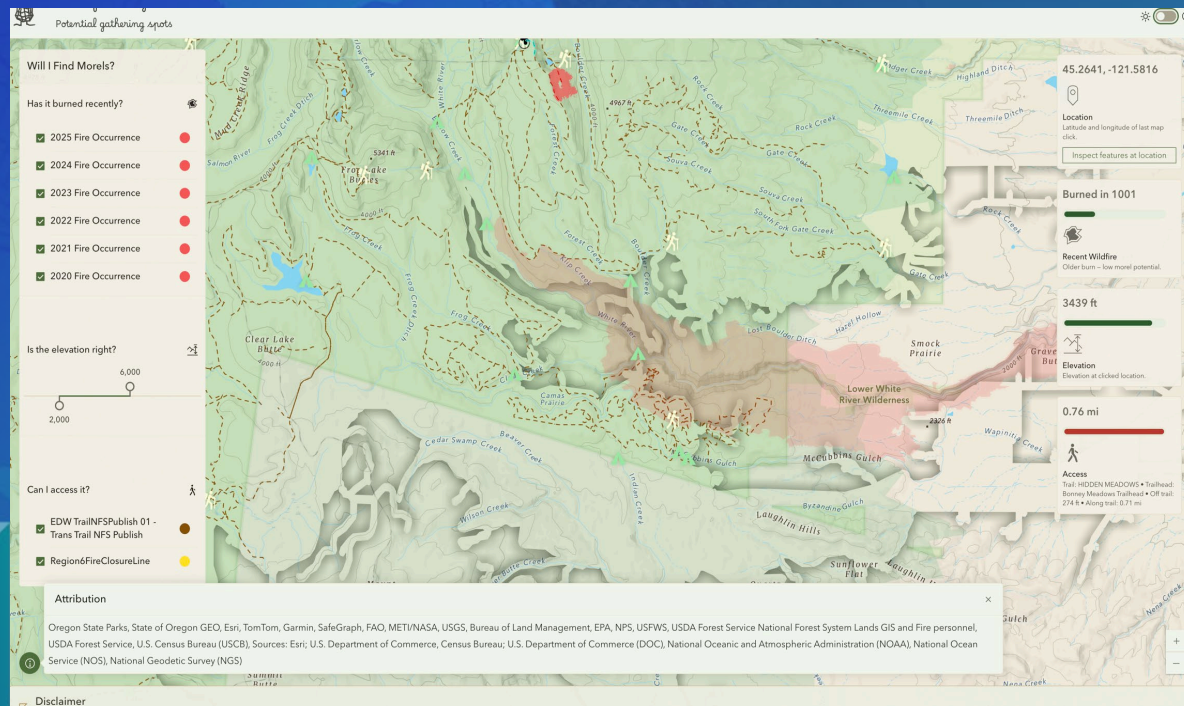
Name	Default	Description
<code>--arcgis-layout-overlay-space-top</code>		Since 4.34 Specifies the top padding for the layout.
<code>--arcgis-layout-overlay-space-bottom</code> 		Since 4.34 Specifies the bottom padding for the layout.
<code>--arcgis-layout-overlay-space-left</code>		Since 4.34 Specifies the left padding for the layout.
<code>--arcgis-layout-overlay-space-right</code>		Since 4.34 Specifies the right padding for the layout.
<code>--arcgis-view-color-focus</code>		Since 5.0 Specifies the focus outline color for the view.

Attribution customization

<https://developers.arcgis.com/javascript/latest/references/map-components/components/arcgis-map/#properties>

Properties

Property	Attribute	Type
<code>allLayerViews</code> readonly		<code>ReadonlyCollection<LayerView></code>
<code>analyses</code>		<code>Collection<Analysis></code>
<code>animationsDisabled</code>	<code>animations-disabled</code>	<code>boolean</code>
<code>aria</code>		<code>MapView["aria"]</code>
<code>attributionItems</code> readonly		<code>MapView["attributionItems"]</code>
<code>attributionMode</code>	<code>attribution-mode</code>	<code>"dark" "light" null undefined</code>
<code>autoDestroyDisabled</code>	<code>auto-destroy-disabled</code>	<code>boolean</code>
<code>ground</code>	<code>ground</code>	<code>Ground</code>
<code>hideAttribution</code>	<code>hide-attribution</code>	<code>boolean</code>
<code>highlights</code>		<code>Collection<HighlightOptions></code>
<code>interacting</code> readonly		<code>boolean</code>
<code>ipsInfo</code>		<code>IPSInfo null undefined</code>
<code>itemId</code> reflected	<code>item-id</code>	<code>null undefined string</code>
<code>layerViews</code> readonly		<code>Collection<LayerView></code>



Demo:

Custom Attribution

René Rubalcava

Discussion and resources



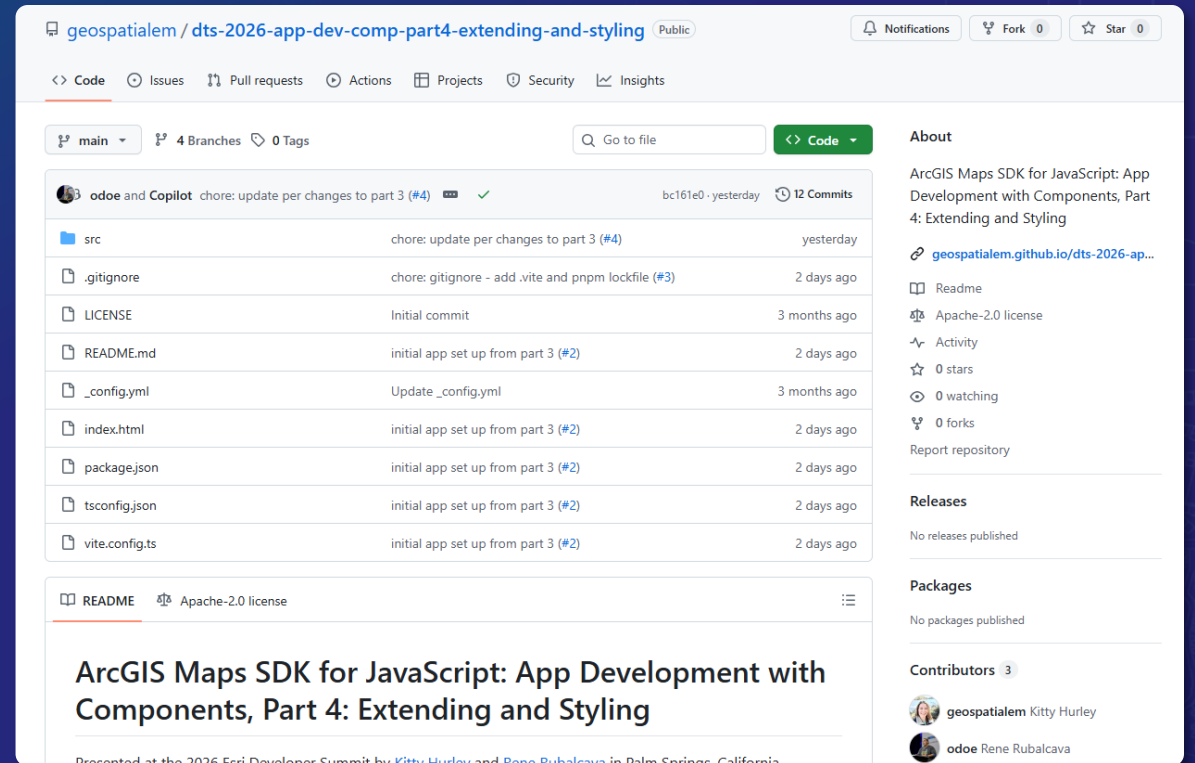
Accessibility considerations

- Provide high color contrast
 - Ensure contrast ratios are upheld
 - Light and dark modes
- Choose a readable font
 - Font size
 - Line height
- Ensure the focus is visible
 - Sufficient color contrast
- Offer multiple customization configurations
 - Disable animations if reduced motion is set
 - Provide higher contrast modes if Windows high contrast mode is enabled



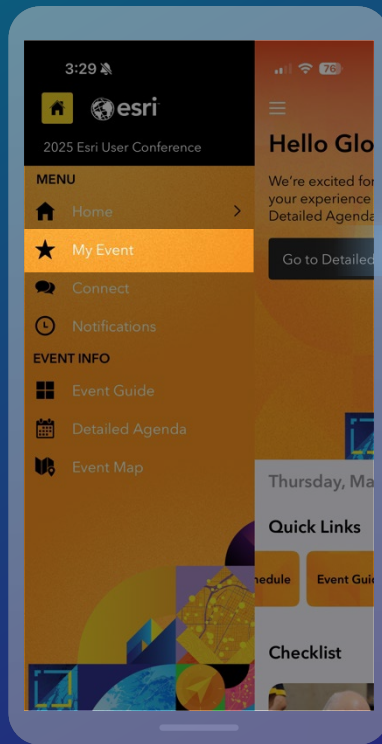
Resources

- GitHub repository
 - <https://esriurl.com/dts26-extend-styling>
- Calcite design tokens
 - <https://esriurl.com/calcite-tokens>
- Build a UI with JavaScript Maps SDK
 - <https://esriurl.com/js-build-ui>

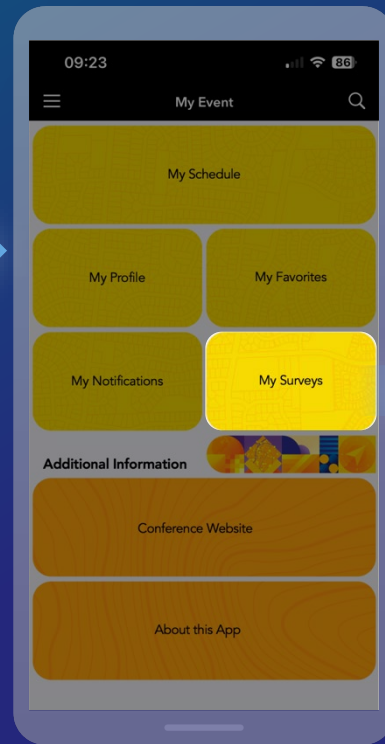


Please share your feedback in the app

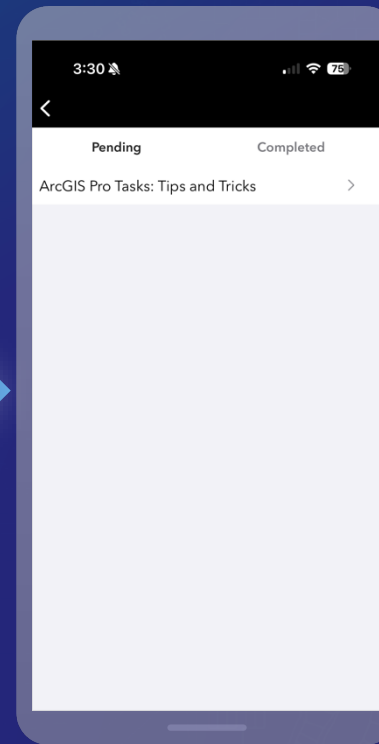
Select “My Event” in the event app main menu



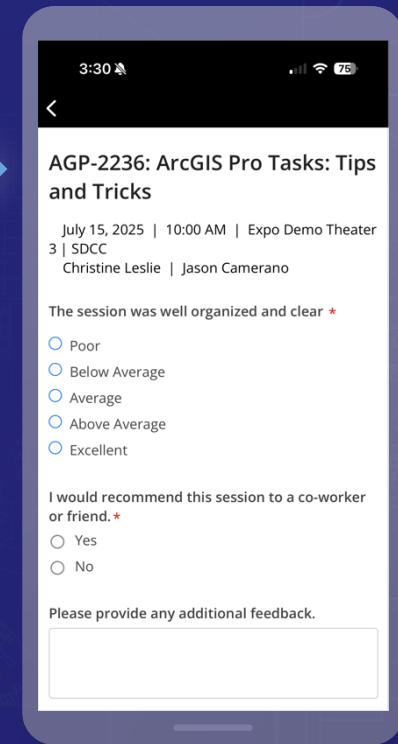
Select “My Surveys”



Select the desired survey



Complete the survey





```
map = new Map(BasemapStyle.ArcGISNavigation(),  
    initialViewpoint = new Viewpoint(40.72,  
    -73.995,  
    72000));
```

```
mapView.Map = map;
```

```
struct ContentView: View {
```

```
    @State private var map = {  
        let map = Map(  
            basemapStyle: .arcGISNavigation  
        )
```

```
        map.initialViewpoint = Viewpoint(  
            latitude: 40.72,  
            longitude: -73.995,  
            scale: 72_000  
        )
```

```
        return map
```

```
    }
```

```
    var body: some View {  
        MapView(map: map)
```

```
    }
```



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SCIENCE
OF
WHERE®